

Car Parking

SWE-4202 — Object-Oriented Programming — Lab Final

Language: Java (JDK 17)

Input: standard input

Output: standard output

1. The Car Park

The basement car park of a shopping complex in Dhaka charges its customers by the hour. Write a program that keeps the park for one day and answers questions about it. The business facts below are the specification; the emphasised phrases are load-bearing.

1. The park is a row of slots of **three different kinds** — bike slots, regular slots and large slots — and it has **its own count of each kind**, given on the first line of the input. Those counts never change.
2. **Every fee in this car park belongs to the kind of slot, not to the kind of vehicle.** A slot charges one price for the first hour and another for every hour after it, and it charges a **surcharge, once**, whenever it is holding a vehicle that belongs in a different kind of slot. Table 1 gives all of them. A motorcycle left standing in a regular slot is therefore billed at the regular slot's rates and pays the regular slot's surcharge — exactly as anything else in that slot would be.
3. **A motorcycle, a car and a truck are all vehicles**, and what distinguishes them is **which kinds of slot they will accept, and in what order** (Table 2). A vehicle takes **the first free slot in its own order**: a motorcycle takes a bike slot if one is free, else a regular slot, else a large one; a car takes a regular slot if one is free, else a large one; a truck takes a large slot or nothing. **The first kind in a vehicle's order is where it belongs**; standing in any of the others costs that slot's surcharge.
4. Every vehicle has a **number plate, fixed at the moment it enters**. It enters having stood **no hours at all**, and its hours **only ever grow, as time passes in the park**. Nothing may set them to a value of its choosing. **However short a stay is, it is billed as one whole hour**.
5. Time passes **for the whole park at once**: the input says how many hours have gone by, and every vehicle standing in the park has stood that much longer.
6. **No vehicle may stay longer than the maximum stay**, given on the second line of the input. The moment a vehicle's hours reach it, the vehicle is **evicted at once, without being asked**, and its slot is free again immediately.
7. **Being removed costs more than leaving**. An evicted vehicle is billed then and there for **the whole of the maximum stay** — never for the hours beyond it that the clock may already have run, so the bill does not depend on how large a jump in time caused the eviction — and then for **a removal charge of one further hour for every ten hours of the maximum stay, or part of ten**. A maximum stay of ten hours is billed as eleven; one of eleven hours is billed as thirteen. The removal charge is counted **in hours**, so like every other hour it costs whatever the slot the vehicle is standing in charges for a further hour.
8. A discount scheme **is not a kind of vehicle**; it is something a vehicle **carries**. Every vehicle carries exactly one scheme, named when it enters, and the scheme applies to **the whole bill, surcharge included**. Three schemes are in use today (Table 3) and the management invents a new one before every festival, so **a fourth scheme must not require any change to any vehicle or to any slot**. **A discount may never reduce a bill below zero**; a bill of zero is a perfectly ordinary bill.
9. The park keeps a running total of what it has **earned**. A bill is earned **only when its vehicle actually leaves the park** — whether it drives away or is evicted. Asking what a vehicle owes must not add anything to the total, and neither must admitting one.

10. A vehicle that has left, for either reason, is **no longer in the park**: its slot is free again and it can no longer be asked about.

Kind of slot	First hour (BDT)	Each further hour (BDT)	Surcharge (BDT)
Bike slot	10	5	0
Regular slot	30	20	15
Large slot	50	40	25

Table 1: Every fee in the park. A vehicle that has stood h hours in a slot owes first + $(h - 1) \times$ further, plus that slot's surcharge if the vehicle belongs in another kind of slot, and then its own discount.

Vehicle	Written as	Slots it will accept, best first
Motorcycle	BIKE	bike slot, then regular slot, then large slot
Car	CAR	regular slot, then large slot
Truck	TRUCK	large slot only

Table 2: The only thing that distinguishes the three kinds of vehicle.

Scheme	Written as	Effect on a bill of b Taka
No discount	NONE	b (unchanged)
Student	STUDENT	$b - \lfloor 20b/100 \rfloor$ (twenty percent off, rounded down)
Weekend	WEEKEND	$\max(0, b - 10)$ (a flat ten Taka off)

Table 3: The three discount schemes.

2. Refusals

Three things can go wrong. Each of them is an error, and **each must be reported by throwing an exception** of a class you define yourself; your program then catches it, counts it where the input asks for a count, and carries on with the next line. A refusal never changes the park: no slot is taken, no vehicle is removed, nothing is earned.

1. A vehicle arrives with **no plate** — its entry slip was left blank.
2. A vehicle arrives and **no slot it will accept is free**. Note that the park can be full for a truck and have room for a motorcycle at the very same moment.
3. A plate that is **not in the park** is asked about, or told to leave.

An eviction is not a refusal: it is the park working as designed, and it must never be counted as one.

3. Input

Input is a sequence of lines. Every line is one instruction: a word in capitals, then fields separated by single spaces. A plate is a single word without spaces; a plate written as a single dash - means the entry slip was left blank. Every number is a whole number and arrives as text. The first line is always SLOTS, the second is always MAXSTAY, the last line is always END, and there is nothing else in the input.

Line	Meaning
SLOTS <bike> <regular> <large>	The park has that many bike slots, regular slots and large slots. The first line of the input.
MAXSTAY <hours>	No vehicle may stand in the park for that many hours or more. The second line of the input.
BIKE <plate> <scheme>	A motorcycle arrives and tries to park. <scheme> is NONE, STUDENT or WEEKEND.
CAR <plate> <scheme>	A car arrives and tries to park.
TRUCK <plate> <scheme>	A truck arrives and tries to park.
PASSTIME <hours>	That many hours go by. Every vehicle in the park has stood that much longer, and any that reaches the maximum stay is evicted and billed. <hours> is always at least 1.
LEAVE <plate>	That vehicle tries to drive away and pay.
BILL <plate>	Question: what does that vehicle owe as things stand?
SLOT <plate>	Question: which kind of slot is that vehicle standing in?
FREE <kind>	Question: how many slots of that kind are free? <kind> is BIKE, REGULAR or LARGE.
COUNT	Question: how many vehicles are in the park now?
EARNED	Question: how much has the park earned so far today?
REFUSED	Question: how many instructions have been refused so far today?
END	The last line.

No two vehicles in the park at the same time carry the same plate. A plate may enter again after the vehicle that carried it has gone.

4. Output

Print exactly one line for each BILL, SLOT, FREE, COUNT, EARNED and REFUSED line, in the order those lines appear, and nothing else.

- For FREE, COUNT, EARNED and REFUSED, print a single whole number, with no currency symbol, no unit and no separators.
- For BILL, print the whole number of Taka the vehicle owes — or the word NONE, in capitals, if no vehicle with that plate is in the park.
- For SLOT, print BIKE, REGULAR or LARGE — or NONE if no vehicle with that plate is in the park.
- SLOTS, MAXSTAY, BIKE, CAR, TRUCK, PASSTIME and LEAVE print nothing at all, whether they succeed or are refused. An eviction prints nothing of its own.

Print no prompts, no headings, no blank lines and no explanatory text.

5. A Sample Run

The lines on the left, typed in that order, produce exactly the lines on the right — thirty-six questions are asked, so thirty-six lines are printed. The park here has **one bike slot, two regular slots and one large slot**, and a maximum stay of **ten hours**.

Input

SLOTS 1 2 1
 MAXSTAY 10
 BIKE DH-11-2233 NONE
 BIKE DH-11-0009 NONE
 CAR DH-14-5678 STUDENT
 CAR DH-14-9000 NONE
 TRUCK CT-19-0001 NONE
 SLOT DH-11-2233
 SLOT DH-11-0009
 SLOT DH-14-5678
 SLOT DH-14-9000
 FREE BIKE
 FREE REGULAR
 FREE LARGE
 COUNT
 REFUSED
 BILL DH-11-2233
 BILL DH-11-0009
 BILL DH-14-5678
 BILL DH-14-9000
 PASSTIME 3
 BILL DH-11-2233
 BILL DH-11-0009
 BILL DH-14-5678
 BILL DH-14-9000
 LEAVE DH-11-0009
 FREE REGULAR
 EARNED
 COUNT
 PASSTIME 8
 COUNT
 EARNED
 FREE BIKE
 FREE REGULAR
 FREE LARGE
 BILL DH-11-2233
 SLOT DH-14-9000
 BIKE DH-11-0001 WEEKEND
 SLOT DH-11-0001
 BILL DH-11-0001
 PASSTIME 2
 BILL DH-11-0001
 CAR - NONE
 LEAVE XX-99-9999
 BILL XX-99-9999
 REFUSED
 TRUCK CT-19-0002 NONE
 SLOT CT-19-0002
 BILL CT-19-0002
 LEAVE DH-11-0001
 EARNED
 COUNT
 END

Output

BIKE
 REGULAR
 REGULAR
 LARGE
 0
 0
 0
 4
 1
 10
 45
 24
 75
 20
 85
 56
 155
 1
 85
 3
 0
 804
 1
 2
 1
 NONE
 NONE
 BIKE
 0
 5
 NONE
 3
 LARGE
 50
 809
 1

5.1 Where every answer comes from

#	Question	Answer	Why
1	SLOT DH-11-2233	BIKE	The first motorcycle finds the one bike slot free.
2	SLOT DH-11-0009	REGULAR	The second finds it taken and falls back to a regular slot.
3	SLOT DH-14-5678	REGULAR	The first car takes the other regular slot.
4	SLOT DH-14-9000	LARGE	No regular slot is left, so the second car falls back to the large one. The truck that follows is turned away: nothing it accepts is free.
5	FREE BIKE	0	
6	FREE REGULAR	0	The bike in a regular slot took one of these.
7	FREE LARGE	0	
8	COUNT	4	Four vehicles in four slots of three kinds.
9	REFUSED	1	The truck, and only the truck.
10	BILL DH-11-2233	10	Bike slot, no hours stood yet, billed as one hour: 10.
11	BILL DH-11-0009	45	Regular slot: 30, plus that slot's surcharge of 15, because a motorcycle belongs in a bike slot.
12	BILL DH-14-5678	24	Regular slot: 30, no surcharge (a car belongs there), less twenty percent.
13	BILL DH-14-9000	75	Large slot: 50, plus its surcharge of 25, because a car belongs in a regular slot.
14	BILL DH-11-2233	20	Three hours have passed: $10 + 2 \times 5$.
15	BILL DH-11-0009	85	$30 + 2 \times 20 + 15$. Same three hours, different slot, different price.
16	BILL DH-14-5678	56	$30 + 2 \times 20 = 70$, less twenty percent.
17	BILL DH-14-9000	155	$50 + 2 \times 40 + 25$.
18	FREE REGULAR	1	The motorcycle in a regular slot has driven away.
19	EARNED	85	Its 85 is earned at its departure, and nothing was earned before it.
20	COUNT	3	
21	COUNT	0	Eight more hours puts all three at eleven, past the ten-hour maximum, so all three are evicted where they stand.
22	EARNED	804	Each is billed for the ten-hour stay plus one removal hour — eleven — and not for the eleventh hour the clock actually ran: bike slot $10 + 10 \times 5 = 60$; regular slot $30 + 10 \times 20 = 230$, less twenty percent, 184; large slot $50 + 10 \times 40 + 25 = 475$. With the earlier 85 that is 804.
23	FREE BIKE	1	
24	FREE REGULAR	2	Every slot is free again, of every kind.
25	FREE LARGE	1	
26	BILL DH-11-2233	NONE	Evicted, so no longer in the park.
27	SLOT DH-14-9000	NONE	The same for the car that stood in the large slot.
28	SLOT DH-11-0001	BIKE	A new motorcycle, and the bike slot is free again.
29	BILL DH-11-0001	0	One hour in a bike slot is 10, and the weekend scheme takes a flat 10 off. Not -0 , and not refused.
30	BILL DH-11-0001	5	Two hours: $10 + 5 = 15$, less the flat 10.
31	BILL XX-99-9999	NONE	A plate that never entered. The blank plate before it and the attempt to make it leave were both refused.
32	REFUSED	3	The truck, then the blank plate, then that departure.
33	SLOT CT-19-0002	LARGE	A truck at last: the large slot has been free since the eviction.
34	BILL CT-19-0002	50	One hour in a large slot, and no surcharge — a truck belongs there.
35	EARNED	809	The motorcycle's 5 is added when it leaves. The truck's own 50 is not: the truck is still parked.
36	COUNT	1	The truck.

Questions 11 and 15 are where the fees show whose they are: the same motorcycle, in the same three hours, costs 20 in a bike slot and 85 in a regular one. Question 22 is the eviction: the stay itself never billed beyond the maximum, the removal hour added on top, and the whole of it earned without

anybody asking. Question 35 is the till: a bill exists long before it is earned.

6. How Your Program Is Marked

Your program reads its instructions from the console. They are **typed in by hand**, one line at a time, and your program prints its answer to a question as soon as that question is typed. It must not read a file, must not expect anything on the command line, and must not print anything of its own — no prompts, no headings, no “Enter a command” — because every line it prints is compared against the line that was expected.

Marking is done by typing in **several sets of instructions you have not seen** and comparing what your program prints with what it should have printed, line for line. Each set exercises the rules of Sections 1 and 2 in a different combination, with different slot counts and a different maximum stay. **The sample in Section 5 is not the examination:** it uses every rule at least once, but a program that answers it perfectly can still lose marks on the sets used for marking. A program that does not compile scores nothing, so hand in something that compiles even if it does not answer every question.

7. Constraints

1. **Java 17**, standard library only. `ArrayList`, `List`, `HashMap` and the rest of `java.util` are allowed.
2. **Integer arithmetic only.** No `double` and no `float`: every bill is a whole number of Taka, and the rounding in Table 3 is exactly what Java’s integer division already does.
3. **Hours are whole hours** and the clock only moves forward.